DATA ITEM DESCRIPTION

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Public reporting burden for this collection of information is estimated to average 110 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, D.C. 20503.

1. TITLE	2. IDENTIFICATION NUMBER
Electromagnetic Environmental Effects (E3) Verification Procedures (E3VP)	DI-EMCS-81541

3. DESCRIPTION/PURPOSE

3.1 The E3VP describes the methods of test, analysis, and inspection used by the contractor to verify compliance with the electromagnetic environmental effects (E3) interface and performance requirements of a system.

3.2 The E3VP provides the means for the government to understand and duplicate verification methods used by the contractor to verify E3 requirements.

4. APPROVAL DATE *(YYMMDD)*	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
970318	F-11		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for data resulting from the work task described by 4.1 of MIL-STD-464.

7.2 This DID is intended for airborne, sea, space, and ground systems, including associated ordnance.

7.3 This DID is normally applied to the engineering and manufacturing development phase of a program, but it can be used in any phase.

7.4 This DID is related to DI-EMCS-81540 and DI-EMCS-81542.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A7254

10. PREPARATION INSTRUCTIONS

10.1 Format. Contractor format is acceptable.

10.2 Content. The E3VP shall describe the overall verification methods and shall provide detailed verification procedures (test, analysis, and inspection, as applicable) for each E3 requirement specified in the contract for the system being developed.

10.2.1 Summary information. The procedures shall summarize the following:

10.2.1.1 Introduction/Background.

(Continued on page 2)

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Block 10, Preparation Instructions (Continued)

a. System description, including any pertinent information regarding verification issues.
b. Statement of any assumptions and limitations associated with verification.
c. General objectives.

10.2.1.2 Scope. General description of overall verification matrix being used to demonstrate compliance with requirements, including the relative role of analyses, tests and inspections.

10.2.1.3 Methods of verification. Abstracts of the procedures used for verifying each E3 requirement listed in section 10.2.2 below.

10.2.1.4 Engineering factors. Any important engineering factors affecting the verification procedures, such as facilities, resources, safety, reports, and security.

10.2.2 Detailed information. The E3VP shall provide detailed technical information covering the overall verification methodology (audit trail of various analyses, tests, and inspections and their interrelationships) used to verify compliance for each of the interface requirement areas listed below addressed in contractually imposed requirements. The E3VP shall include detailed procedures (analyses, tests, and inspections, as applicable) for each area, including the types of information listed in the following subsections.

a. Margins.
b. Intrasystem electromagnetic compatibility (EMC), including where applicable: ship hull intermodulation interference, internal electromagnetic environments, powerline transients, and multipaction.
c. Intersystem EMC.
d. Lightning.
e. Electromagnetic pulse.
f. Subsystem and equipment Electromagnetic Interference (EMI), including where applicable: non-developmental items, commercial items, electromagnetic spectrum compatibility, and DC magnetics.
g. Electrostatic charge control, including where applicable: vertical lift and in-flight refueling, precipitation static, and explosive subsystems.
h. Electromagnetic radiation hazards, including where applicable: hazards of electromagnetic radiation to personnel, hazards of electromagnetic radiation to fuel, and hazards of electromagnetic radiation to ordnance.
i. Life cycle E3 hardness.
j. Electrical bonding, including where applicable: power current return path, antenna installations bonding, and EMI bonding.
k External grounds, including where applicable: aircraft grounding jacks.
l. TEMPEST.
m. Emissions control.
n. Electronic protection.

10.2.2.1 Scope.

a. Objective of verification for the particular area.
b. References.

10.2.2.2 Verification article.

a. Identification of the physical configuration, such as structural features, mechanical and electrical equipment installed, and software status.
b. Description of system functions (or subsystem/equipment functions) that are required or available.

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c. Description of provisioned equipment (items that are part of the resultant system operation but are not necessarily developed under the contract), such as weapons, pods, and payloads that are required.

d. Operating details of the system.

10.2.2.3 Elements of verification.

a. Models, techniques, and tools used for analysis and predictions and their specific application to this system.

b. Step by step procedures.

c. Determination of applicable margins and the methods to be used for demonstration.

d. Selection of critical circuits, functions, and subsystems.

e. Pass or fail criteria and methods of quantifying and evaluating degradation.

f. Description of test articles, test facilities, test equipment (including instrumentation on and off the system), support equipment, and calibration techniques.

g. Method of simulating operational performance when actual operation is impractical.

10.3 Other information sources. When other information sources contain data required by this DID, these sources shall be referenced rather than being duplicated within these procedures.

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